How to: Keycaps

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Table Of Contents

1. Introduction 2
2. Blank Keycaps 3
3. *Keyboard* Layouts 4
4. Laser *Engraving* 5
5. Filled *Laser* Engraving 9
6. UV-Decals 10
   - 6.1. General Information about UV-decals 10
   - 6.2. Wear testing 11
   - 6.3. Manual Alignment 12
   - 6.4. The Process of applying UV Decals 13
     - 6.4.1. Required Tools 13
     - 6.4.2. Cleaning the Blank Keycap 13
     - 6.4.3. Step One: Cutting Out the Decal 14
     - 6.4.4. Step Two: Positioning the Decal 14
     - 6.4.5. Step Three: Attaching the UV Decal 14
     - 6.4.6. Step Three: Peeling Off the Top Foil 15
7. UV Printing 16
8. Dye Sublimation 18
   - 8.1. Description of Dye Sublimation 18
   - 8.2. Wear testing 20
   - 8.3. Hot Plate Method for Dye Sublimation 21
     - 8.3.1. Building the mold 21
     - 8.3.2. Cutting The Transfer Paper 22
     - 8.3.3. Attaching The Legends to The Keycap 23
     - 8.3.4. Heat Pressing Keycaps 24
     - 8.3.5. Transfer Paper Residues 25
   - 8.4. Dye Sublimation – What Can Go Wrong? 27
9. Comparison of Methods 29
10. Other Ways 30
    - 10.1. Buying Keycaps 30
    - 10.2. Using Original Keycaps 31
    - 10.3. Print Services 31
11. Acknowledgements 32

12. Product References and Disclaimer 32
13. Revision history 32
13.1. Rev. 0.0 32

1. Introduction

For every retro-computer mechanical keyboard project, **keycaps have been a significant challenge**. There are, however, several methods for making decent-looking keycaps. Making keycaps is also the subject of many YouTube videos. I have tried some of these methods with varying degrees of success.

Before making any C64 keycaps, there was a major task to complete: **creating the legends with** Inkscape. Inkscape is free and open-source vector graphics software, which I use for almost every project to create case labels, logos and block diagrams.

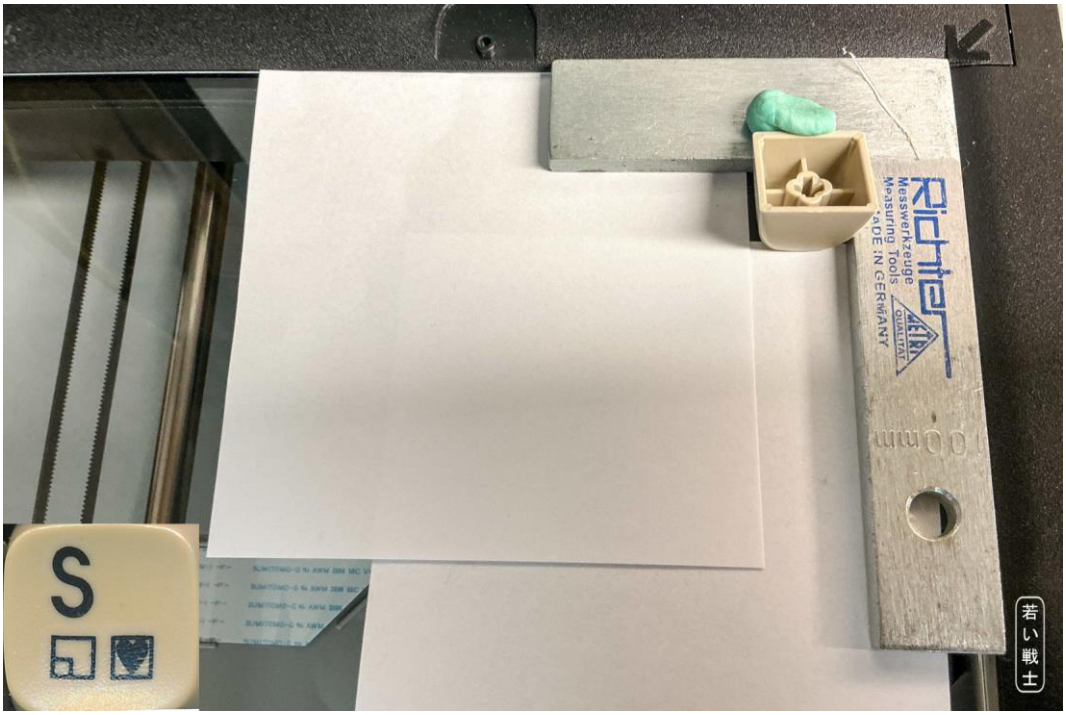


Figure 1: Scanning a single keycap

The legend drawings are **based on scans of the top of every individual keycap of a C64C keyboard**. These keycaps have top-printed legends only. A great help was discovering that the font used for the letters is ITC Avangarde Gothic LT Medium, which is also freely available on GitHub. To use the *.svg (Inkscape) files provided in this repository, you need to install this font first. It needs to be installed **for all users** so that it is available to Inkscape.

The Inkscape **keycap legends** are exported as **pixel-based *.png images** at a resolution of **600 dpi**. These are the files I used for laser-engraving several sets of keycaps.



Of course, you can also use **C64 Ultimate or original C64 keycaps** with this C64 keyboard project.

2. Blank Keycaps

The basis for all DIY keycaps is **blank keycaps**, which are available from various sources. I have only tried purchasing them from AliExpress, so all of my market research is based on this source.

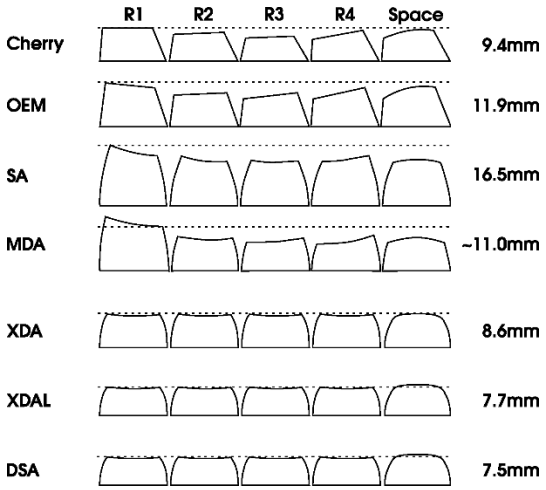


Figure 2: Some keycap profiles

Several keycap profiles are available (Figure 2). The problem is that we do not just need **1U keycaps**, but also **1.5U keycaps** for SHIFT, CONTROL, RESTORE and the function keys, a **2U keycap** for RETURN, and a **7U space bar**—all in the same profile. So far, I have only found a complete set of the required sizes in the **DSA profile**.

Obviously, some keycap profiles have a **different shape for each row**. The original C64 keyboard has **ten different keycap types**: four different 1U keycaps, four different 1.5U keycaps, one 2U keycap, and the space bar.

Most keycaps are made of **PBT** (polybutylene terephthalate), a type of plastic that feels very nice to the touch and is also suitable for **dye-sublimation printing (temperature!)**, unlike ABS. The clear or transparent keycaps I tried were obviously not made of PBT. I cannot recommend clear, transparent or translucent keycaps for dye sublimation or laser marking.



Figure 3: F/J homing keycaps (dye sublimation)

The F and J keys usually have a small raised bar or a bump to help touch typists locate the home-row position without looking at the keyboard. The index fingers rest on these keys when using the standard touch-typing position. Blank **homing keycaps** are somewhat harder to find, but may appeal to keyboard enthusiasts.

3. *Keyboard* Layouts

In addition to the standard keyboard layout, there are **Danish, Swedish, and Spanish keyboard layouts**, with language-specific keycaps. Of course, these also require a different **Kernal ROM**. In the 20th century, the language specific keycaps were modified with stickers - that is what I was told, and it is also what I have seen.



Figure 4: Standard/US keyboard layout

Apart from these, I have heard of a **Turkish keyboard layout**, but I do not have enough information about it. I have also heard of a **German keyboard layout for the SX-64**, but I have no further information about that, **either**.



Figure 5: Danish-specific keycaps on the right-hand side of the keyboard



Figure 6: Swedish- (and Finnish-)specific keycaps on the right-hand side of the keyboard

The Danish and Swedish keyboard layouts require seven specific keycaps each (none of them works for both languages), while Spanish only requires three specific keycaps.

None of the keyboard layouts (even the US-/standard layout) is fully meeting the nowadays used ANSI layouts – sure, there are just more keys today. The Danish Å, Æ, Ø and the Swedish Å, Ö and Ä are in the same position relative to the other letters like on modern keyboards, while the Spanish-specific characters/letters are just moved to three specific keys.



Figure 7: Right-hand side of the Spanish keyboard with the language-specific keycaps

4. Laser *Engraving*

Purchasing a fiber laser just to make a single set of keycaps is not very economical. The **LaserPecker LP3** cost €1,260 in February 2025.

It is important to use a laser that is suitable for engraving plastic keycaps. This requires a **1064 nm infrared laser**. The blue laser, that works for cutting and engraving wood will not produce any good results with plastic/PBT keycaps.



Figure 8: Keycaps made with a fiber laser (LaserPecker LP3)

Actually, the term **"laser engraving"** is somewhat misleading. The keycaps are not really engraved; they are **laser marked**, which means that the surface of the keycaps is locally charred by the laser. The result is a dark grey marking rather than a completely black one, but it still looks good (see Figure 8).

This process works best on **white keycaps**. Lighter colors also work for laser marking, it is required to optimize the parameters for laser marking for each color individually, since different colors react differently to the laser. Darker colors require significantly less energy.

For black/dark plastic, **micro-foaming** can be used to create a light-colored laser marking. The laser locally heats the material, causing microscopic bubbles to form below the surface and producing a contrasting light-grey-brownish mark.



Figure 9: Colored keycaps experiments

It is not possible to **laser-mark transparent or clear keycaps**, as shown in Figure 9. The micro-foaming on the black and grey keycaps has a slight brownish tint. It was also possible to micro-foam the translucent black keycaps (f7/f8, top row, second from the left), but the result did not look good.

Here are some LP3 parameters, that I have used for laser marking the particular colored PBT keycaps. The parameters depend on the keycap color.

Color	Resolution	Power	Depth	Passes
White	4k	40%	1%	4
Grey	4k	70%	3%	4
Orange	4k	30%	1%	4
Black	4k	20%	1%	4
Translucent black	4k	32%	1%	6
Light blue	4k	30%	1%	4
Light violet	4k	35%	1%	4

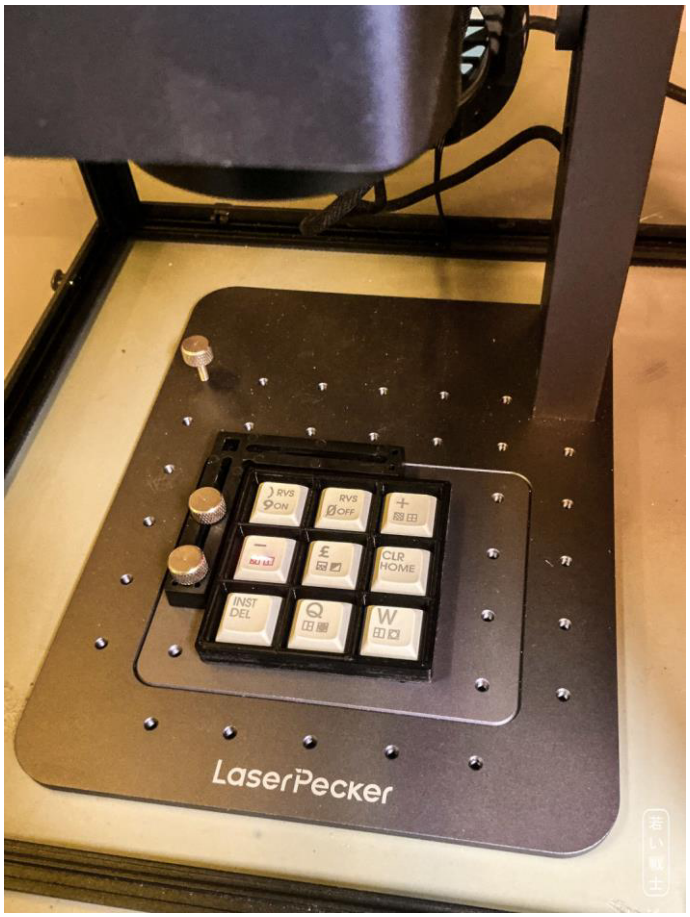


Figure 10: Laser marking nine keycaps simultaneously in a tray

I made the first keycap set **one keycap at a time**. Since positioning the keycap is the most time-consuming part of setting up the marking process, a complete set of 67 keycaps took about **three hours**.

To speed up the process, I made two keycap trays: one that holds **nine 1U keycaps** and another that holds **four 1.5U keycaps** (Figure 11). This significantly reduces the setup time, although it also increases the risk of ruining several keycaps at once—which happened to me once.



Figure 11: both keycap trays loaded with blank keycaps

Find the 3D print files in the \Keycaps\print_trays_3D\ directory.

I have tried different things to dye the laser marking (graphite, crayon, dye sublimation ink), none of that came any close to an acceptable result.

5. Filled *Laser* Engraving

The Legends can be laser engraved about 0.2mm deep onto the keycap and later filled with model paint. I have used Revell matt white for this purpose. johnsander from forum64.de suggested to apply a thick layer over the engraved parts, let it dry for a week and polish the excess away with an abrasive polishing wheel for rotary tools.



Figure 12: Engraved black keycap



Figure 13: Keycap filled with white plastic paint and rotary tool

The white filling looked a bit dirty after polishing and the black surface of the keycap was not perfect, either, so I wiped it with a nano-sponge (my favorite is Ma Ant MY-065 because of the size and density) and isopropanol. The result looks impressive, almost like a double shot keycap. It is quite a bit of work to produce such a keycap. The advantage of this method is, that white on dark is possible. Most likely a coating is required to protect the white legends from getting dirty.

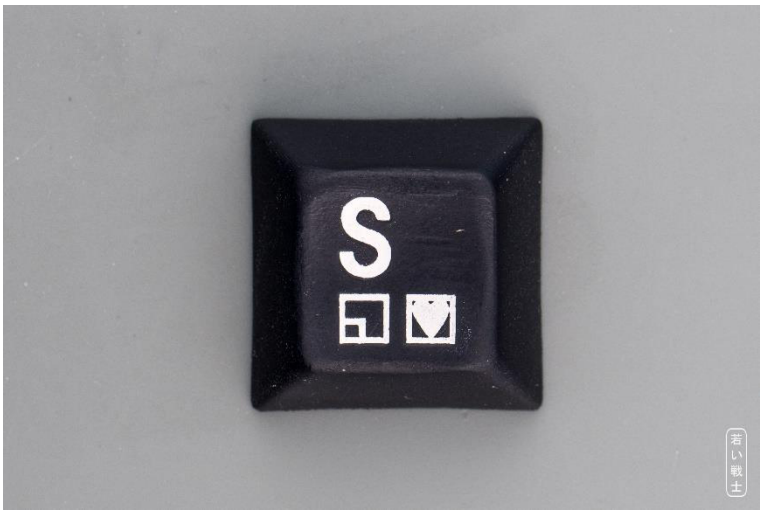


Figure 14: polished keycap

6. UV-Decals

6.1. General Information about UV-decals

During the **documentation phase** of the C64 keyboard project, I had a conversation with the BadgeMan, who provides really great labels and badges for retro-computer restorations and projects. He suggested making **UV decals**, which he then created from the files in this project's repository.



Figure 15: UV decal sheets from BadgeMan

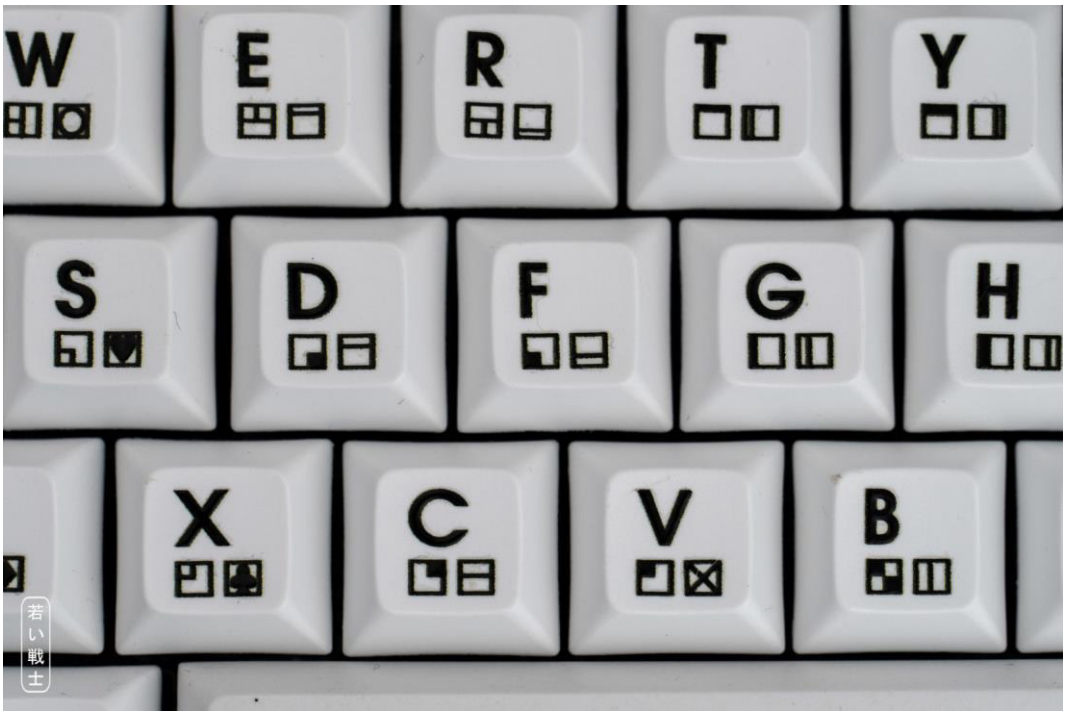


Figure 16: Keycaps with black UV-decals



Figure 17: Keycaps with white UV-decals

6.2. Wear testing

The biggest concern with using UV decals (or any other type of decal) on keycaps was their **durability**. After applying UV decals to a set of keycaps, I therefore tested a RESTORE key that I had deliberately made for destructive testing.

I attached it to a VICE USB keyboard (as a function key), wrote a short BASIC program to count key presses, and then pressed the key **10,000 times**, deliberately applying more friction than would be expected during normal use.

After the counter reached 10,000, I took some pictures of the keycap under a microscope.



Figure 18: RESTORE UV-decal after 10,000 key presses (microscope image)

The result of the test was that I could not see any **significant difference** before and after the test. I therefore do not think that **excessive wear of the UV decals is a concern**.

6.3. Manual Alignment

A further concern was that **manually aligning and positioning the UV decals might be difficult, time-consuming, or result in an inconsistent appearance**.

Mr. BadgeMan told me that I just had to trust my "carpenter's eye." I didn't even know I had such a thing, but I trusted it, and in my opinion, it worked out pretty well. Maybe there is a Dunning–Kruger effect for carpenter's eyes, though.

Applying the UV decals is certainly not something you want to do before breakfast, but it didn't take much longer than an hour and a half to finish the entire set.

What was sometimes difficult and annoying was that, in a few cases, parts of the decals did not stay on the keycap when I peeled off the protective foil. The left PETSCII icon on the C key in Figure 17 was one such case. Carefully reapplying the decal and applying some mild, but not excessive, pressure with a rubber eraser helped in the vast majority of cases, making the damage virtually invisible.



Figure 19: perspective view on a UV decal

6.4. The Process of applying UV Decals

6.4.1. Required Tools

There are actually no fancy tools required for applying UV decals. I only use an X-Acto knife (scalpel) and a pair of scissors. A rubber eraser is helpful for pressing the UV decal onto the keycap.

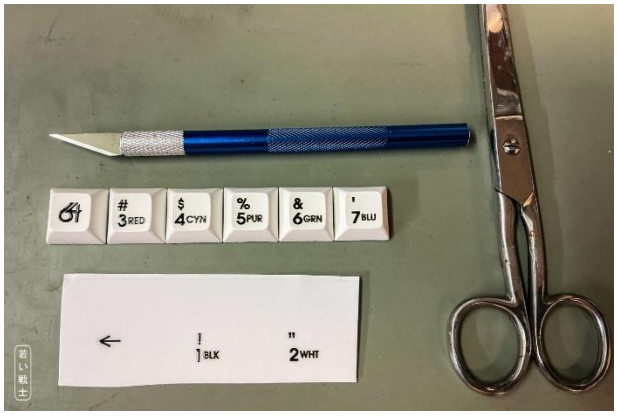


Figure 20: The required tools

6.4.2. Cleaning the Blank Keycap

A good practice is to rub the keycap with isopropanol and a clean, lint-free cloth. I recently came across “Cleanroom Wipers” from the MECHANIC brand on AliExpress, which I really like. I usually use them for cleaning up flux residues (also with the help of isopropanol), and they do an excellent job. They are anti-static, BTW.



Figure 21: Cleanroom whipes

6.4.3. Step One: Cutting Out the Decal

The UV decals are intentionally printed with slightly more space between them. It is important to cut them out generously enough to leave sufficient space to hold them. This is essential for positioning the decal above the keycap before applying it.

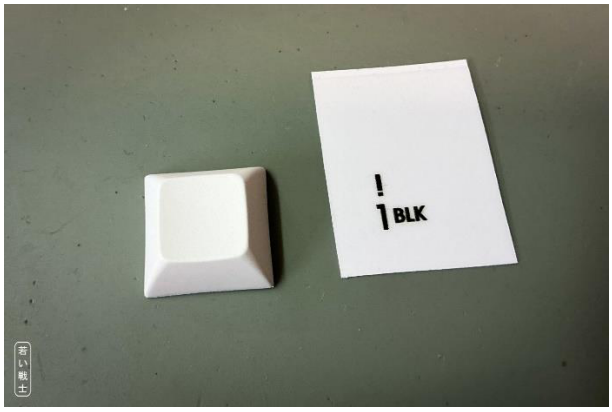


Figure 22: cutting out the decal

6.4.4. Step Two: Positioning the Decal

After peeling off the protective paper, it is easiest to hold and position the decal with the X-Acto knife. I usually hold the opposite corner with my fingers, because a slight tension helps to maneuver the decal above the keycap.

Some decals are positioned off-center, so it is a good idea to keep a picture of the keyboard layout at hand (Figure 4). Eyeballing the correct position of the decals works remarkably well. Most keycap legends are centered, and the PETSCII symbols on many keycaps provide a good reference point. The horizontal positioning should not be completely off, but most key rows are not vertically aligned anyway, so there is definitely some tolerance. The function keys are actually not difficult either—or perhaps you simply have some practice by the time you get there.

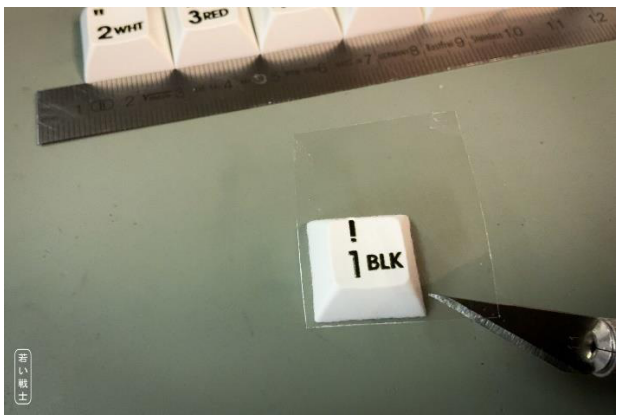


Figure 23: Positioning the decal

6.4.5. Step Three: Attaching the UV Decal

After sticking the decal onto the keycap, I briefly press it onto the keycap with a rubber eraser. Not for very long, and definitely not very hard—just using the weight of my hand. We don't want to deform or break it.

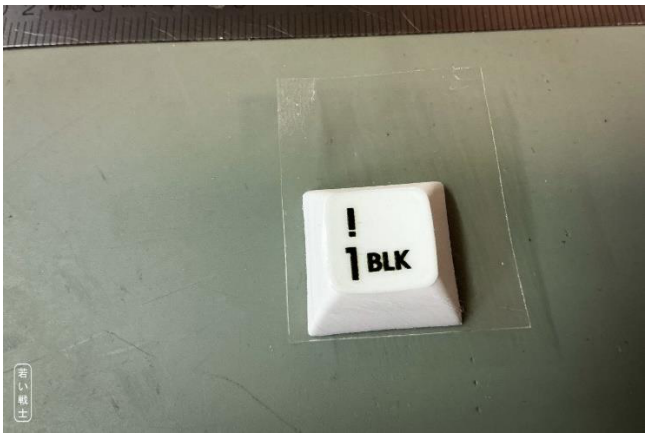


Figure 24: Sticking the UV decal to the keycap

6.4.6. Step Three: Peeling Off the Top Foil

Peeling off the foil is a crucial step. You need to carefully watch the keycap legend to prevent it from breaking or losing parts of it (like the dot of the "!"). **If something breaks, not everything is lost**. Just lower the foil partially back onto the keycap, press it lightly onto the keycap, and maybe try peeling it off again from a different direction. This sounds more dramatic than it actually is. I only had problems with a handful of keycaps, and I was able to fix all of them except for the aforementioned dot of the "!".

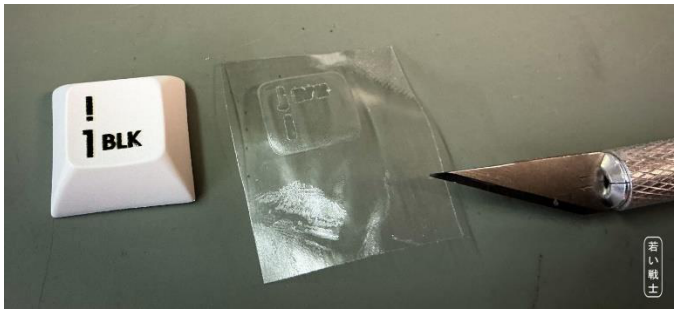


Figure 25: Peeling off the foil

7. UV Printing

UV printing is the method that requires the highest initial investment. UV ink is expensive and has a limited shelf life: it expires after about a year, or after only three months once opened (Mimaki). UV printers start at around €2,500 and can cost considerably more. There have also been a couple of crowdfunding campaigns recently (as of late 2026).

UV printing cannot really be recommended for producing just one or a few keyboards at home. There are UV printing services available, though. Since I do not do UV printing myself, I cannot provide any further insight into this method.



Figure 26: UV print (white on black keycaps) (by AKCZ/norths.de)

As with UV decals, it is possible to have white legends on black (or other dark) keycaps. The result may look slightly unsharp, although it is somewhat better than with white UV decals. When printing light colors onto dark objects, a white ink layer is typically used underneath the colors to provide a sufficiently bright and opaque base. The additional print layer can result in slightly less sharp edges, which may be noticeable with small keycap legends.

There is a single printing tray that can hold all keycaps (except the space bar, which does not require a print). This tray is meant for UV printing. The size is 191.3mm x 162.2mm x 7.9mm. The print file (vector graphics) is \Keycaps\Inkscape\DSA_keycaps_in_single_tray.svg. Find the 3D print files in the \Keycaps\print_trays_3D directory.



Figure 27: Tray for UV printing, which holds all keycaps

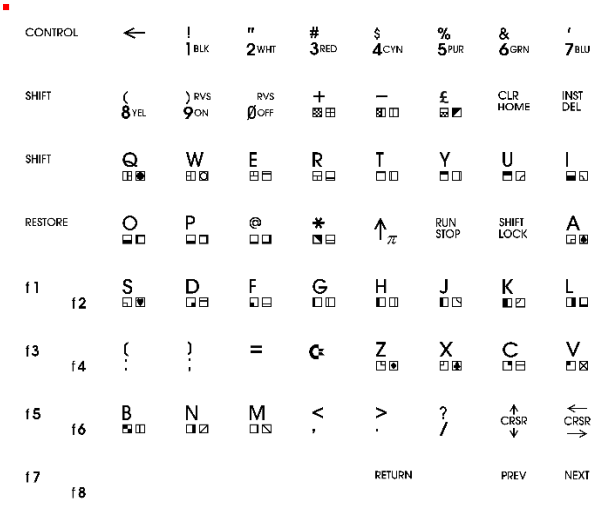


Figure 28: vector graphic file suitable for the above tray

8. Dye Sublimation

8.1. Description of Dye Sublimation

Dye sublimation is quite a popular method for adding legends to keycaps. The legends are printed in mirror image onto special transfer paper using a dye-sublimation inkjet printer and dye-sublimation ink. Colored prints are also possible, but only on white keycaps.



Figure 29: Hand-made dye sublimation keycaps

The paper with the printed legend is cut out and placed on the keycap with the ink side facing down. The paper is secured with heat-resistant Kapton tape. **Heat (about 200 °C) and pressure are then applied, ensuring that the paper is held closely against the keycap surface.** The dye sublimates and **diffuses into the keycap material**.

Sublimation is when a solid (dye) turns directly into a gas without becoming liquid first. Due to the process, legends are slightly less sharp, than laser marked legends. This is a minimal imperfection and hardly noticeable.

The keycap material should be **PBT rather than ABS**, as PBT can withstand the temperatures required for the process.

The dye sublimation transfer paper print can be found in the /Keycaps/Inkscape folder. It is not mirrored, since the printer driver of my dye sublimation printer driver already mirrors the printout. The sheet contains every legend twice to allow failures. Further, there are the language specific keys for Spanish, Danish and Swedish.

If you can buy pre-printed dye sublimation transfer paper (see Figure 31), the investment in equipment is comparatively small. My method uses a 10 cm × 10 cm **hot plate** (about €27 on AliExpress), some **aluminum foil** for making a mold, and 15mm wide **Kapton tape**.



Figure 30: Colored dye sublimation keycaps



Figure 31: Dye sublimation transfer paper with C64 keycap legends (mirrored)

I have found videos suggesting the use of a **hair straightener** with adjustable temperature and **silicone blocks** for heat transfer. The adjustable hair straighteners I have seen start at about €50, and silicone is not particularly cheap either. In addition, silicone does not conduct heat very well, whereas the hot plate can also be used for SMD soldering, for example, when assembling the keyboard PCB. Hot

plate soldering is a very good method for soldering single-sided SMD PCBs. I have been using this method for four years now and have never gone back to other methods.

Further tools:

- A benchtop dispenser for Kapton tape
- An X-Acto knife with a fresh blade
- A steel ruler
- Tweezers for handling the transfer paper cutouts
- A (self-healing) cutting mat
- Pliers that can reach into the keycap from below or a 5.5m – 7mm socket wrench
- A pair of scissors
- A count down timer

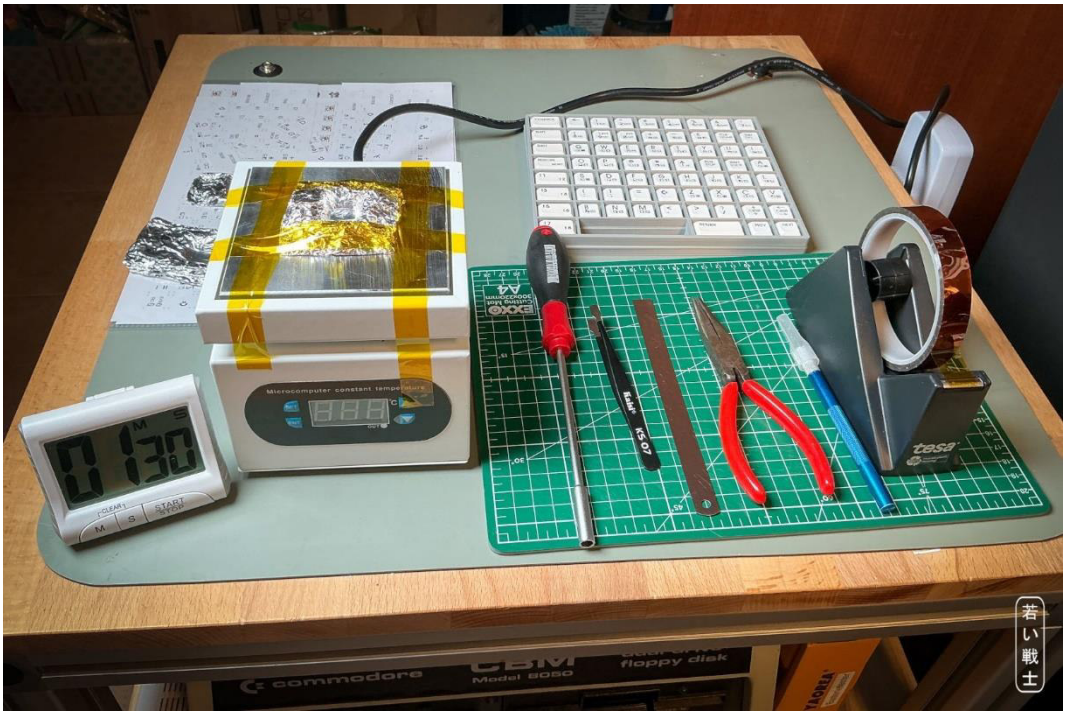


Figure 32: Workbench for manual dye sublimation

8.2. Wear testing

Keycaps with dye-sublimated legends can withstand a reasonable amount of wear, since the dye diffuses into the plastic. I rubbed a discarded keycap two hundred times with a fiberglass pen, which I normally use for cleaning electrical contacts. After the test, the dye showed no visible signs of damage, and the keycap legend looked as good as it did before the test.

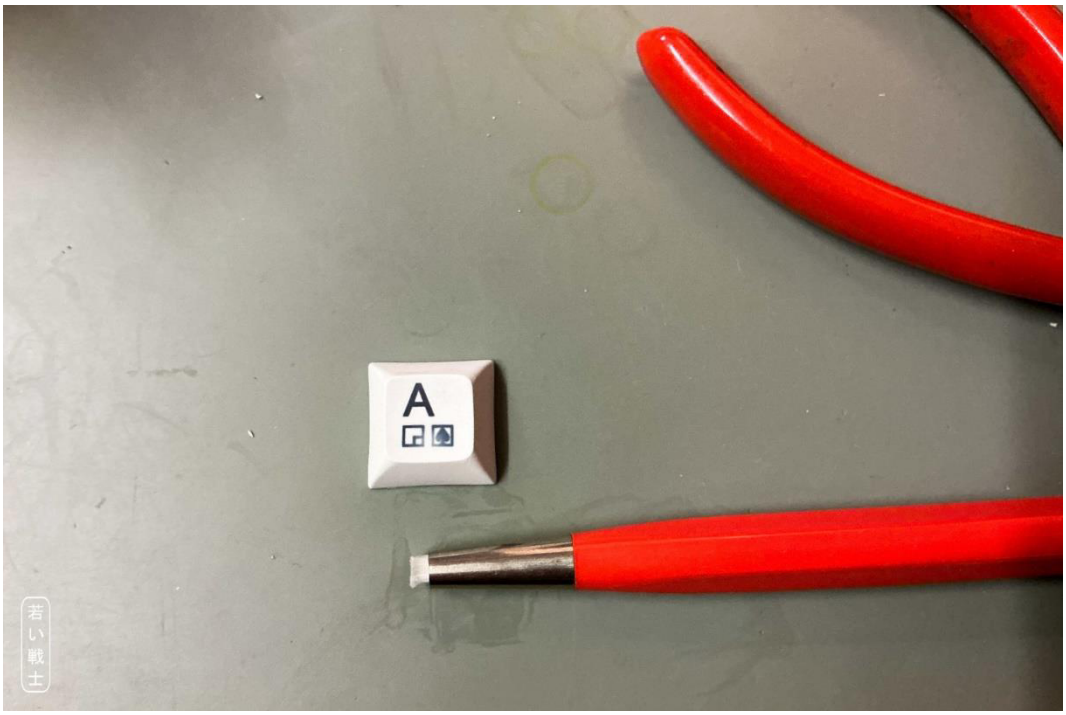


Figure 33: The (discarded) "A" keycap before the wear test (with glass fiber pen)

As an additional durability test, the keycaps were soaked in isopropanol for about 12 hours. They did not lose any sharpness or contrast.

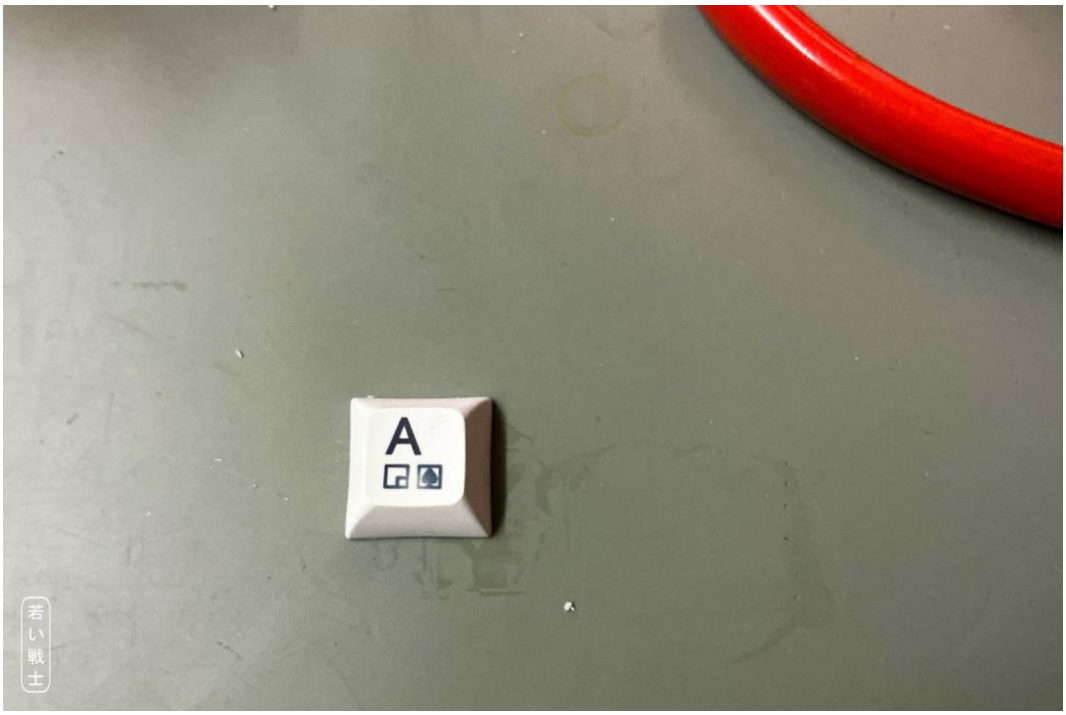


Figure 34: Keycap after wear testing

8.3. Hot Plate Method for Dye Sublimation

8.3.1. Building the mold

It is important that heat and pressure are applied evenly across the entire top surface of the keycap. Aluminum foil can be shaped to fit the cavity of this surface and also conducts heat very well.

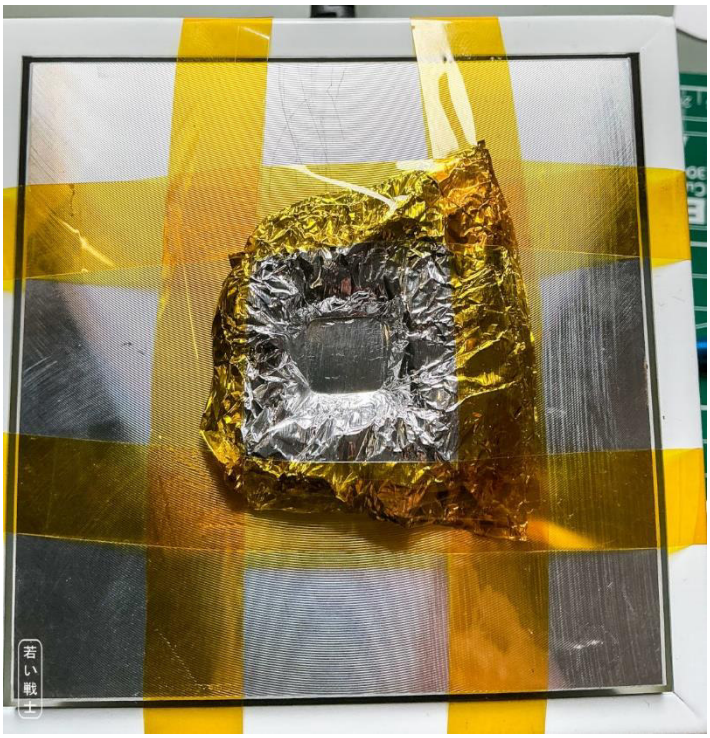


Figure 35: Aluminum foil mold attached to the hot plate with Kapton tape

Start with a generous amount of aluminum foil and form it into a square several times larger than the keycap. Some extra material should be placed underneath the area that will be shaped to fit the keycap. Then, firmly press a keycap of the size to be printed firmly onto the aluminum foil to form the mold. A few small wrinkles in the foil are not a major problem, but a dent can result in partially blurred legends.

Finally, secure the mold to the hot plate with (heat-resistant) Kapton tape.

8.3.2. Cutting The Transfer Paper

I added four register marks to the keycap legends to make it easier to cut them precisely to the size of the keycap's top surface. The register marks are shaped like crosses. This allows you to cut an entire row of keycaps while leaving a small part of each cross visible, so you can still identify the vertical cut lines.

The cuts need to be made quite precisely, as this makes it easier to properly position the individual pieces of transfer paper on the keycap.

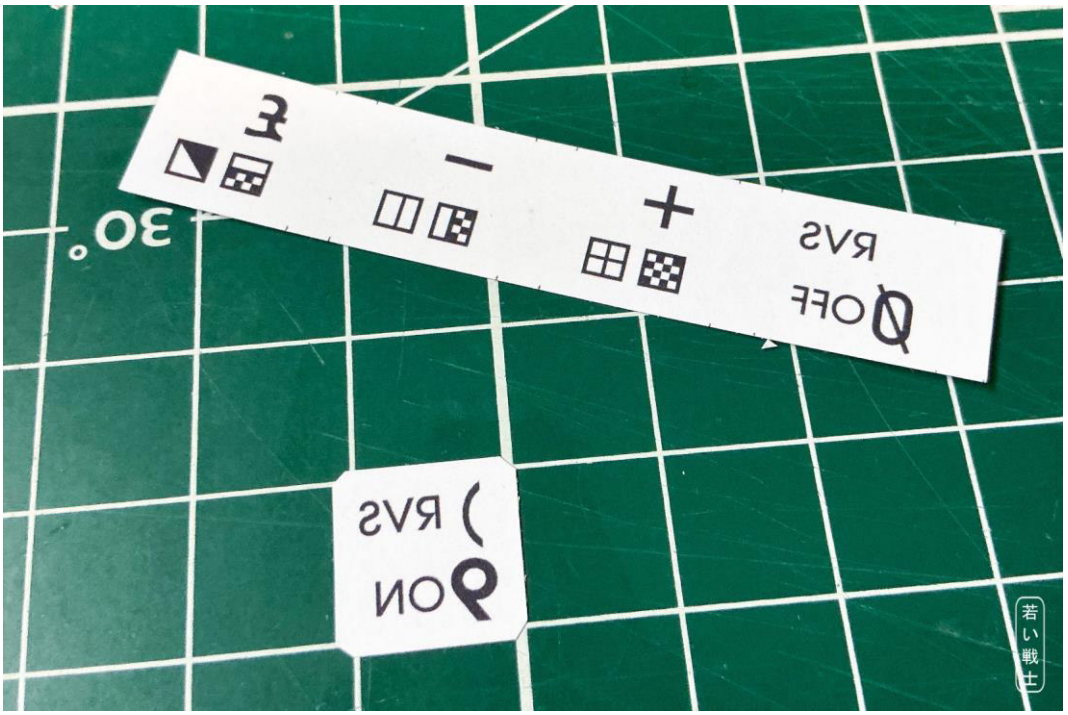


Figure 36: Cutting out a single keycap legend

It is important to cut off all four corners of the legend, as they may contain a small part of a register mark. Otherwise, you may end up with some wonderful little dots on your keycap that will not wear off. (see Figure 43).

8.3.3. Attaching The Legends to The Keycap

First, cut a sufficiently long piece of Kapton tape. For a 1U keycap, about 4 cm is sufficient; larger keycaps require a slightly longer piece. The tape should be long enough to fully cover the sides of the keycap.

Place the tape on the workbench with the sticky side facing up. Then place the piece of transfer paper, printed side up, centered on the tape. I use tweezers for this job.

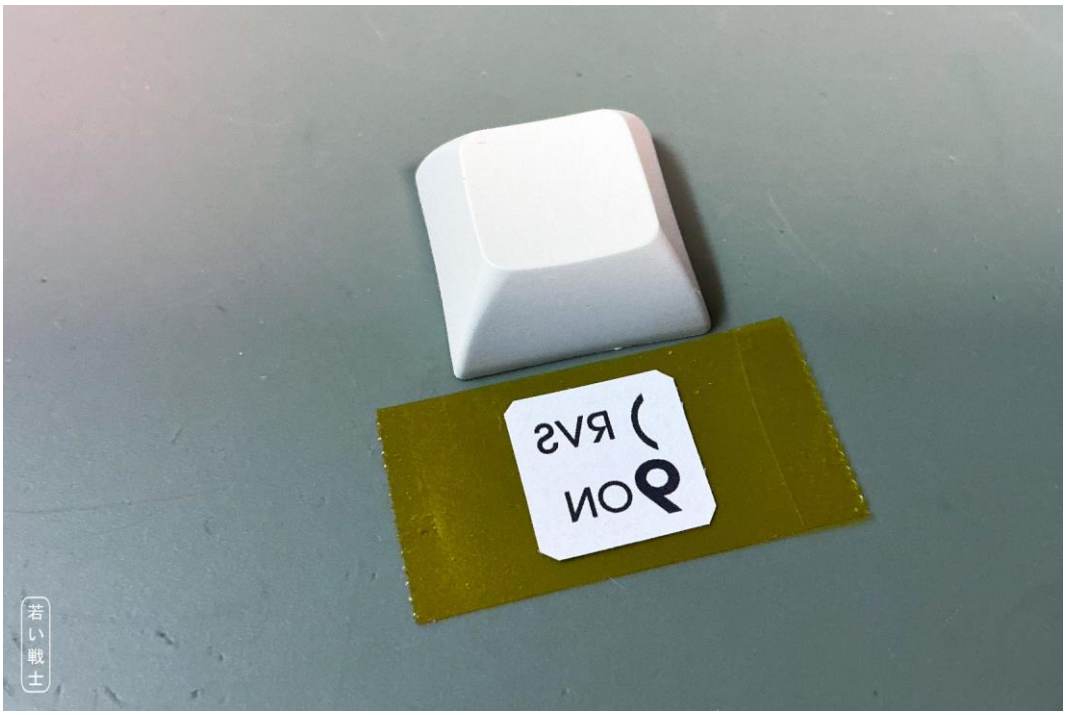


Figure 37: Placing the transfer paper on the Kapton tape



Figure 38: Attaching the transfer paper on the keycap

Since the cut transfer paper pieces are sufficiently precise, **positioning the legends is quite straightforward**. After adjusting the position, **press the tape onto the top** of the keycap and **rub it down on both sides** so that the legend stays in place. The paper should be able to reach all the way into the deepest part of the keycap's top cavity.

8.3.4. Heat Pressing Keycaps

As mentioned before, the important factors in dye sublimation are heat (200°C), time (90 seconds), and pressure.

Set the hot plate to 200°C. Let it heat up to that temperature, then wait a few more minutes until the aluminum foil mold has also reached the required temperature.

Hold the **keycap with pliers directly above the mold**. Start the **90-second countdown**, then **lower the keycap precisely into the mold** and **press down firmly** (Figure 39). If you do not position it accurately enough, you may damage the mold. This is the point where **concentration is required**. There is no need to rush it—taking an extra five seconds to position the keycap correctly does not matter.

Note: **I have damaged a keycap** due to pressing very hard. The pliers deformed the top surface. I then switched to pressing with a socket wrench (as can be seen in Figure 32).

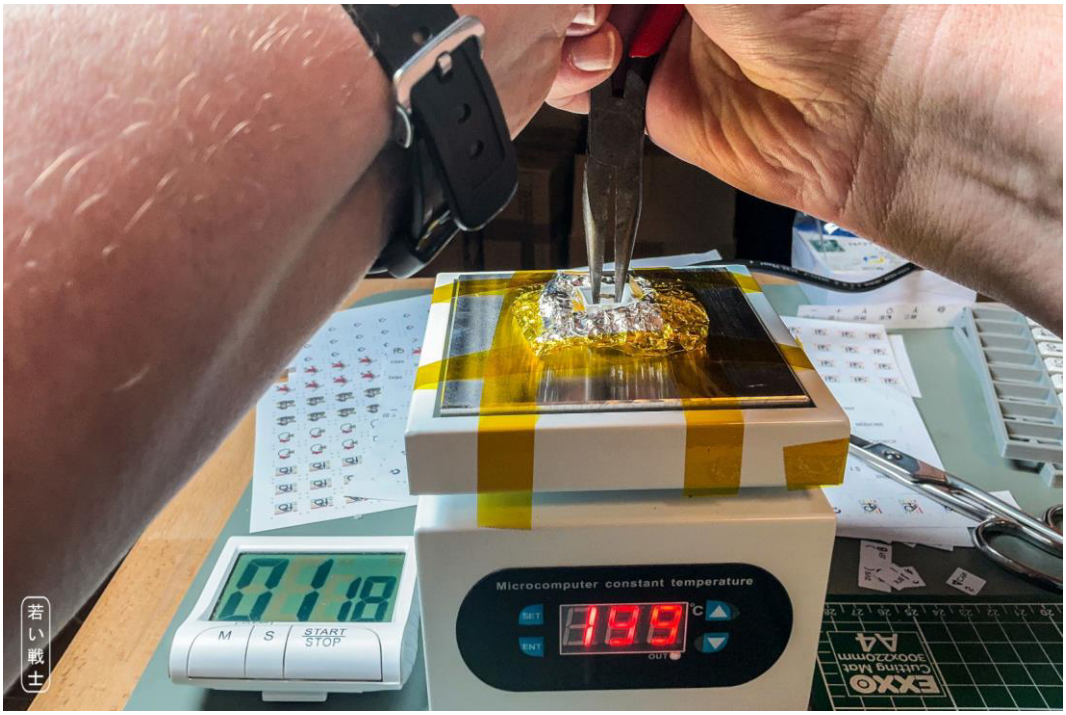


Figure 39: Heat pressing the keycap

I measured the force I apply using a scale. It showed more than 11 kg, which corresponds to a force of about 110 N. This might be my typical "more power" approach to things.

Before I tried holding the keycap with pliers, I used tweezers, which allowed me to apply a force of about 70 N (equivalent to 7 kg on the scale). This sometimes resulted in blurred legends and discarded keycaps. Also, **you can lose your grip on tweezers more easily than on pliers**. Keep in mind that the hot plate is at 200°C.

After increasing the applied force, I stopped ruining keycaps.

8.3.5. Transfer Paper Residues

After heat pressing, significantly less dye remains on the transfer paper than before (Figure 40). The high pressure may cause the transfer paper to stick to the keycap. Usually, it can be peeled off using an X-Acto knife. Slide the blade between the paper and the keycap. There is usually at least one side where it is easier to get the blade underneath.

If small pieces of paper remain stuck to the keycap (see Figure 41), soaking them in hot, soapy water or running them through a dishwasher cycle may help remove them.

I have not tried the dishwasher method, but I have placed the keycaps in a heat-resistant bowl, added some dish soap, and poured boiling water over them. Once the water had cooled to lukewarm, I rubbed the remaining paper off with my fingers.



Figure 40: Keycap and transfer paper after heat pressing



Figure 41: Residues of transfer paper after heat pressing

8.4. Dye Sublimation – What Can Go Wrong?

I have not made many sets of dye-sublimated keycaps yet. After a short learning phase to figure out how to make them, I did not have many failed attempts—in fact, only one.



Figure 42: Failes dye sublimation attempts

- **Left arrow:** Pressing time too short
- **PREV/FREEZE:** Pressing time too short
- **£:** Insufficient and uneven pressure
- **A:** A dent in the aluminum foil mold between the PETSCII characters
- **RETURN:** Keycap deformed by excessive pressure from overly pointed pliers
- **SHIFT LOCK:** Obviously not made of PBT
- **NEXT/MENU:** Excessive temperature (230°C)

Initially, I did not use a timer, which resulted in the two keycaps with insufficient pressing times. The dye simply needs enough time to transfer to the keycap. A pressing time of 90 seconds turned out to be sufficient.

At first, I pressed the keycaps down using tweezers, but the pressure I could apply (about 70 N) turned out to be insufficient. Areas subjected to too little pressure resulted in blurred legends. Using pliers and, later, a socket wrench, I could easily apply about 110 N (equivalent to the weight of an 11 kg mass). This was sufficient even for the larger 1.5U keycaps and the large 2U RETURN keycap.

My first attempt at dye-sublimating the RETURN keycap failed because I applied too much pressure with overly pointed pliers. The pliers almost pierced the keycap material.

On one occasion, I accidentally dropped a keycap onto the aluminum foil mold, leaving a dent in its surface. The corresponding area of the legend was not sharp.

The clear keycap was declared to be made of PBT, but it obviously was not. After doing some research, I doubt that clear or transparent PBT keycaps are readily available at all. I would recommend avoiding keycaps described as “clear,” “transparent,” or “translucent” for dye sublimation.

I deliberately applied excessive heat (230°C), and the result was as expected. The recommended hot plate temperature is 200°C.

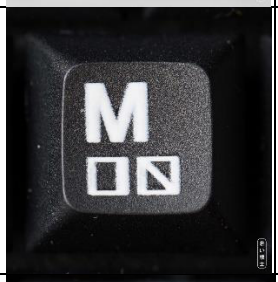
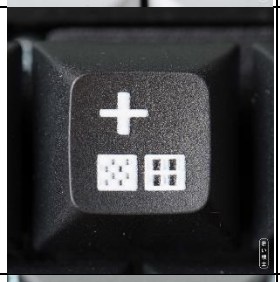
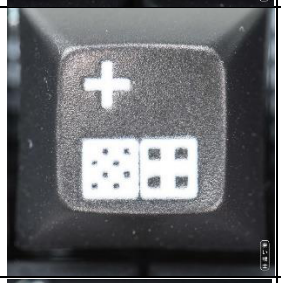


Figure 43: The register marks for cutting made their way to the keycap

Here, I forgot to cut off the corners of the transfer paper. They were barely noticeable on the paper but stand out clearly on the keycap.

9. Comparison of Methods

This table shows the same keycaps produced with the different methods.

Laser Marking				
UV decals (black)				
UV decals (white)				
UV print (white)				
Dye Sub				

Note: The photos were taken on different days, so the camera and lighting setup was not always the same. This results in slightly different sizes and appearances.

The white UV print was made using a different keycap print file. The PETSCII characters are slightly larger, and the letters use a different font (see the “Q”).

Find the original images (not scaled down) in the \Keycaps\pictures\ folder.

	Laser Marking	UV Decals	UV Printing	Dye Sublimation
Costs for a printing service	?	👍	👍	?
Investment for own tools	1600€	(>5000€)	>2500€	500€/30€***
Time per keycap set	1h	1h	?	4h
Costs per print (set)	👍👍👍	👎👎	👎	👍👍
Print sharpness (black on white)	👍👍👍	👍👍	?	👍👍
Print contrast (black on white)	👍👍	👍👍👍	?	👍👍👍
Prints white on black	👎	👍	👍	👎👎👎
Print sharpness (white on black)	👍👍👍	👍	👍👍	d.n.a.
Print contrast (white on black)	👎 Brownish tone	👍👍👍	👍👍👍	d.n.a.
Multi-color keycaps	👎👎👎	👍👍👍	👍👍👍	👍👍*
Keycaps of different colors	👎	👍👍👍	👍👍👍	👍👍**

d.n.a. = does not apply

* works on white keycaps only

** not too dark, no white dye is possible

*** The investment depends on whether you buy a dye-sublimation printer or use pre-printed transfer paper.

10. Other Ways

10.1. Buying Keycaps

The C64 Ultimate/Blingboard64 keycaps will fit this keyboard project. They are the best possible reproduction of the original keycaps and can be purchased fro. There are “original” and “Blingboard64” keycaps. The latter are the proper choice. They fit the switches in the SWxxA position (modern) and require a 7U stabilizer for the spacebar.

The only disadvantage is that the “PREV” and “NEXT” key cannot be populated, which means, that quite a bit of functionality (Kernal switching, scrolling through the LED strip illumination modes) cannot be used.

10.2. Using Original Keycaps

Using original keycaps is also an option. Since the switch connector of the keycaps does not fit on the MX compatible switches, there is are adapters required for the keycaps. These are the same like for the MechBoard64. They can be purchased from https://www.retrofuzion.com/ (MechBoard 64 Injection Molded Keycap Adapters). If you chose this way, please make sure, that they are not sold out.



Figure 44: 1.5U keycap with keycap adapters (Retro Fuzion)

The disadvantage is again the 9U spacebar is occupying the space for the "PREV" and "NEXT" key, which offer the Kernal switching and LED strip functionality. Also, the adapters add some height to the keycaps (1.9mm).

The original 9U space bar requires a 9U stabilizer, which cannot be purchased. The instructions on making such a stabilizer are part of this repository.

The 1.5U switches have to be populated in the proper position (SWxx instead of SWxxA) for these keycaps.

10.3. Print Services

I have tried norths.de for UV printing, I did not pay money for that, though. So, I cannot tell the price of the service. I am evaluating the UV printing capabilities of teiledrucker.at, who is specialized on printing plastic bricks. The keycaps are not printed yet, due to a printer breakdown. I will take pictures and update this document ASAP.

In forum64.de, I have received recommendations for other keycap services.

- https://yuzukeycaps.com/de
- https://maxkeyboard.com/

I did not try them out, so I cannot say anything about them.

1. Check if they only offer standard keyboard layouts. The C64 keyboard is not standard
2. Check the keycap profiles, they offer. If those are different for every row, make sure they offer 1.5U for every row, and 2U keycaps for Row 3 (see Figure 2)
3. Ask for the price.
4. Please let me know your experience.

11. Acknowledgements

I would like to thank everyone who contributed ideas and suggestions for making keycaps for this project:

- **Marco Van de Meulenhof** for the idea of using UV decals and for printing them.
- **johnsander** from forum64.de for explaining how engraved keycaps can be filled with paint.
- **Anthony Jacoway** for the idea of using aluminum foil as a mold for dye-sublimation printing on keycaps. This was a game changer!
- **David Ludwig (norths.de)** for UV-printing the set of black and orange keycaps.
- **Peter Schulz (teiledrucker.at)** for discussing the keycap printing tray with me.

I am also grateful to everyone who followed my progress in making the keycaps, asked questions, and joined the discussions. Your interest and feedback have helped me put together this documentation and may well inspire further keyboard and keycap projects.

12. Product References and Disclaimer

Throughout this documentation, I mention specific **products**, **brand names**, and **suppliers**. These references are provided **solely for informational purposes** and are not intended as advertising or endorsements. I have not received any compensation or other benefits for mentioning them. I refer to these products and suppliers simply because I have used them myself and can therefore share my personal experience with them.

13. Revision history

13.1. Rev. 0.0

- Initial release